ELECTRICAL MACHINES · CH2

Power-factor correction, 3-phase, equivalent-circuit tests

Cheat sheet · Day 02 · Sun 25 Oct · pack M2

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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POWER TRIANGLE   S = V I* = P + jQ = S∠-φ   P = S cosφ   Q = S sin(-φ)   pf = P/S
  φ = current angle wrt voltage: lag φ<0, Q>0 (L) | lead φ>0, Q<0 (C). Sign of Q fror
  Parallel loads: add P's, add Q's (never S's).

PFC (C in parallel; P never changes)
  1 P, Q_old   2 Q_S = +P tan(acos pf) lag / -P tan(...) lead / 0 unity
  3 Q_C = Q_S - Q_old (<0)   4 C = |Q_C|/(ω V²), V across C, check 50 vs 60 Hz
  5 S_new = P/pf, I_new = S_new/V.  Reverse: Q_C = -ωCV², Q_S = Q_old + Q_C
  One pf -> two C values (under/over-corrected). Over-correction -> leading, I rises

3-PHASE (assume star; ratings = line V, line I, total P)
  Star: VL = √3 V1, IL = I1      Delta: VL = V1, IL = √3 I1
  S = 3V1I1 = √3 VL IL;  P = √3 VL IL cosφ;  pf(3ph) = pf(1 phase)
  Same Z: star -> delta on same supply: I1, V1 ×√3; IL, P, Q, S ×3; pf same
  Cap bank, |Q_C|/3 each:  C_Y = |Q_C|/(ω VL²)  [rated VL/√3]
                           C_Δ = |Q_C|/(3 ω VL²) [rated VL]     C_Y = 3 C_Δ

BATTERY   V = E - I Rs;  E = Voc;  Rs = (Voc - V)/I = Voc/Isc
          loss = I² Rs;  Pmax = E²/(4Rs) at RL = Rs (η = 50%)

V,I,P TESTS   S = VI, Q = +√(S²-P²)
  series: R = P/I², X = Q/I²     parallel: R = V²/P, X = V²/Q
  L = X/ω, C = 1/(ω|X|). 3ph: per phase first (V1 = VL/√3, P1 = P/3).
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